



T1-3S Module Datasheet

Version: 20260320

Online Version

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T1-3S module is a Wi-Fi and Bluetooth Low Energy (LE) combo module developed by Tuya Smart. It is composed of a highly integrated radio-frequency identification (RFID) chip T1 and a few peripheral components. It can run in the station (STA) mode and access point (AP) mode as well as support connection over Bluetooth LE.

1. Overview

T1-3S has a built-in 32-bit MCU with a maximum running speed of 160 MHz, built-in 2 MB flash memory, and 288 KB RAM. The module supports cloud connectivity, and the MCU's extended instructions for signal processing enable it to efficiently implement audio encoding and decoding. Various peripherals are available, such as PWM, UART, and ADC. Up to five channels of 32-bit PWM output make the chip ideal for high-quality LED control.

1.1. Features

- Built-in low-power 32-bit CPU that also acts as an application processor.
- Clock rate of 160 MHz.
- Operating voltage range: 3.0V to 3.6V.
- Peripherals: 5 PWM pins, 2 UART pins, and 1 ADC pin.
- Wi-Fi connectivity
- IEEE 802.11b/g/n.
- Channels 1 to 14 at 2.4 GHz.
- Support WEP, WPA/WPA2, WPA/WPA2 PSK (AES), and WPA3 security modes.
- The maximum output average power is +17 dBm for IEEE 802.11b transmission.
- Support STA, AP, and STA + AP combo working modes.
- Two pairing modes are supported, namely bluetooth mode and access point (AP) mode. Both modes are suitable for pairing with Android and iOS devices.
- The onboard PCB antenna has a peak gain of 1.68 dBi.
- Operating temperature range: -40°C to 105°C .
- Bluetooth connectivity
- Bluetooth Core Specification v5.2.
- The transmitter power in Bluetooth mode is 6 dBm.
- Integral Wi-Fi and Bluetooth coexistence interface.
- The onboard PCB antenna has a peak gain of 1.68 dBi.

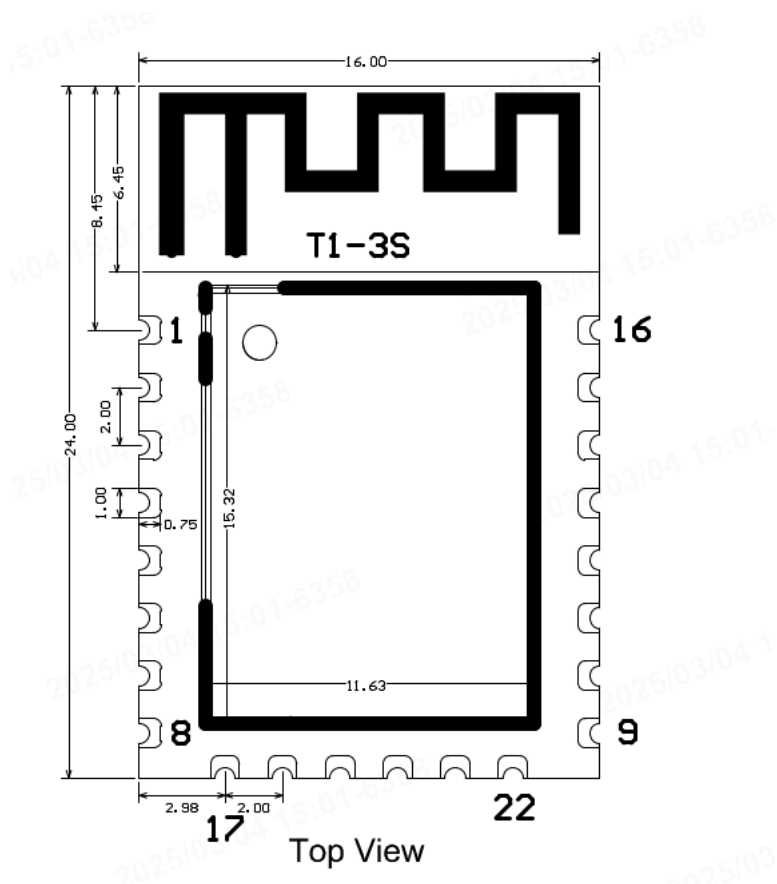
1.2. Scope of applications

- Smart building
- Smart home and electrical appliance
- Smart socket and light
- Industrial wireless control
- Baby monitor
- IP camera
- Smart bus

2. Module interfaces

2.1. Dimensions and footprint

The T1-3S dimensions are 16.0 ± 0.35 mm (W) $\times$ 24.0 ± 0.35 mm (L) $\times$ 2.8 ± 0.15 mm (H). The figure below shows the dimensions of the T1-3S module.



2.2. Pin definition

Pin No.	Symbol	I/O type	Feature
1	RST	I	The low-level reset pin, active high, and pulled up internally, corresponding to CEN on the IC.

Pin No.	Symbol	I/O type	Feature
2	ADC3	I/O	ADC pin, corresponding to P20 on the IC.
3	CEN	I	Enable pin, pulled up internally. When externally pulled low, it can be used as a reset pin.
4	P22	I/O	GPIO_22, corresponding to P22 on the IC.
5	P6	I/O	GPIO_6, corresponding to P6 on the IC. Support hardware PWM0.
6	P8	I/O	GPIO_8, corresponding to P8 on the IC. Support hardware PWM2.
7	P9	I/O	GPIO_9, corresponding to P9 on the IC. Support hardware PWM3.
8	3V3	P	Power supply pin (DC 3.3V).
9	GND	P	Ground pin.

Pin No.	Symbol	I/O type	Feature
10	P23	I/O	GPIO_23, corresponding to P23 on the IC.
11	TX2	I/O	UART2_TXD (used for log printing), corresponding to P0 on the IC.
12	RX2	I/O	UART2_RXD (used for log printing), corresponding to P1 on the IC.
13	P24	I/O	GPIO_24, corresponding to P24 on the IC. Support hardware PWM4.
14	P26	I/O	GPIO_26, corresponding to P26 on the IC. Support hardware PWM5.
15	RX1	I/O	UART1_RXD to receive the user data, corresponding to P10 on the IC.
16	TX1	I/O	UART1_TXD to transmit the user data, corresponding to P11 on the IC.

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- The maximum input level of the ADC port is VBAT (typical supply voltage is 3.3 V). It is recommended to use MΩ level external voltage divider resistors and place a 100 nF ground filter capacitor close to the ADC port.
- P indicates the power pin, and I/O indicates the input and output pin.
- For more information about MCU general integration, see [Hardware Design of T1x Series Modules](#).
- P6, P8, P9, P24, and P26 I/O ports have default power-on glitches. For applications where high-level signals are active, apply appropriate pull-down resistors to maintain a low level.
- The P9 pin cannot be pulled high by an external resistor during power-on initialization. This pin is not recommended for drive circuits that default to active-low operation.
- For descriptions of other pins, please contact Tuya Business for details.

3. Electrical parameters

3.1. Absolute electrical parameters

Parameter	Description	Min value	Max value	Unit
T _s	Storage temperature	−55	125	°C
V _{BAT}	Supply voltage	−0.3	3.6	V
Electrostatic discharge voltage (human body model)	T _{AMB} −25°C	−4	4	kV
Electrostatic discharge voltage (machine model)	T _{AMB} −25°C	−200	200	V

3.2. Normal operating conditions

Parameter	Description	Min value	Typical value	Max value	Unit
T _a	Operating temperature	−40	-	105	°C
V _{BAT}	Supply voltage	3	3.3	3.6	V
V _{OL}	I/O low-level output	V _{SS}	-	V _{SS} + 0.3	V
V _{OH}	I/O high-level output	V _{BAT} − 0.3	-	V _{BAT}	V

Parameter	Description	Min value	Typical value	Max value	Unit
I _{max}	I/O drive current	-	6	20	mA

3.3. Radio frequency (RF) power

Operating status	Mode	Rate	Transmit/Receive power	Average value	Peak (Typical) value	Unit
Transmit	802.11b	11 Mbit/s	+17 dBm	312	365	mA
Transmit	802.11g	54 Mbit/s	+15 dBm	279	332	mA
Transmit	802.11n	MCS7	+14 dBm	265	320	mA
Receive	802.11b	11 Mbit/s	Continuous reception	17	25	mA
Receive	802.11g	54 Mbit/s	Continuous reception	17	25	mA
Receive	802.11n	MCS7	Continuous reception	17	25	mA

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The above reference data is based on a non-signaling mode with a 99% duty cycle.

3.4. Operating current

Working mode	Status (Ta = 25°C)	Average value	Max (Typical) value	Unit
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Working mode	Status (Ta = 25°C)	Average value	Max (Typical) value	Unit
Quick pairing (Bluetooth)	The module is pairing over Bluetooth. The network status indicator blinks slowly.	80	360	mA
Quick pairing (AP)	The module is in AP mode. The network status indicator blinks slowly.	85	360	mA
Connected	The module is connected to the cloud. The network status indicator is steady on.	90	330	mA

Working mode	Status (Ta = 25°C)	Average value	Max (Typical) value	Unit
Weakly connected	The connection between the module and the access point is intermittent. The network status indicator is steady on.	190	380	mA
Disconnected	The module is disconnected from the cloud. The network status indicator is steady off.	67	380	mA
Module disabled	The module's clock enable (CEN) pin is pulled down.	330	-	μA

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The above data module is powered by a DC steady-state power supply at 3.3 V and is based on the Tuya firmware keyy95dt93s3v7qd, with firmware version v1.0.2. Test data may vary slightly for the same firmware due to differences in modules and power supplies.

4. RF parameters

4.1. Basic RF features

Parameter	Description
Operating frequency	2.412 to 2.484 GHz
Wi-Fi standard	IEEE 802.11b/g/n (channels 1–14) • IEEE 802.11b: 1, 2, 5.5, and 11 Mbit/s
Data transmission rate	• IEEE 802.11g: 6, 9, 12, 18, 24, 36, 48, and 54 Mbit/s • IEEE 802.11n: HT20 MCS0-7
Antenna type	PCB antenna

4.2. Wi-Fi transmitter (TX) performance

Parameter	Min value	Typical value	Max value	Unit
RF average output power, 802.11b CCK mode, 11 Mbit/s	-	17	-	dBm
RF average output power, 802.11g OFDM mode, 54 Mbit/s	-	15	-	dBm
RF average output power, 802.11n OFDM mode, MCS7 (HT20)	-	14	-	dBm
Frequency error	-20	-	20	ppm

4.3. Wi-Fi receiver (RX) performance

Parameter	Min value	Typical value	Max value	Unit
PER < 8%, RX sensitivity, 802.11b DSSS mode, 11 Mbit/s	-	-88	-	dBm
PER < 10%, RX sensitivity, 802.11g OFDM mode, 54 Mbit/s	-	-75	-	dBm
PER < 10%, RX sensitivity, 802.11n OFDM mode, MCS7 (HT20)	-	-73	-	dBm
PER < 10%, RX sensitivity, Bluetooth, 1 Mbit/s	-	-96	-	dBm

4.4. Bluetooth TX performance

Parameter	Min value	Typical value	Max value	Unit
Operating frequency	2402	-	2480	MHz
Transmission rate over the air	-	1	-	Mbit/s
Transmission power	-20	6	20	dBm

Parameter	Min value	Typical value	Max value	Unit
Frequency error	-150	-	150	KHz

4.5. Bluetooth RX performance

Parameter	Min value	Typical value	Max value	Unit
RX sensitivity	-	-96	-	dBm
Max RF signal input	-10	-	-	dBm
Intermodulation	-	-	-23	dBm
Co-channel rejection ratio	-	10	-	dB

5. Antenna information

5.1. Antenna type

The T1-3S module uses an onboard PCB antenna.

5.2. Antenna interference reduction

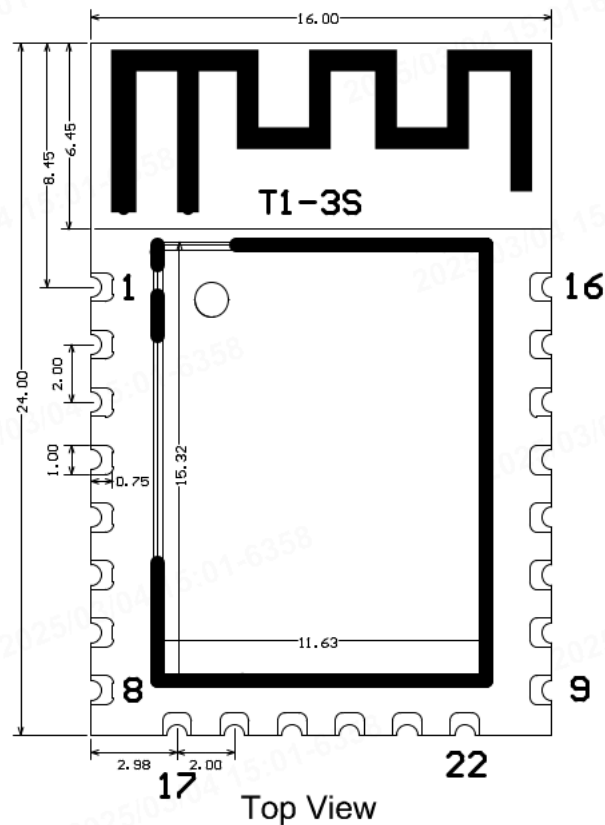
When a PCB antenna is used on a Wi-Fi module, we recommend that the module antenna is at least 15 mm away from other metal components. This can optimize the Wi-Fi performance.

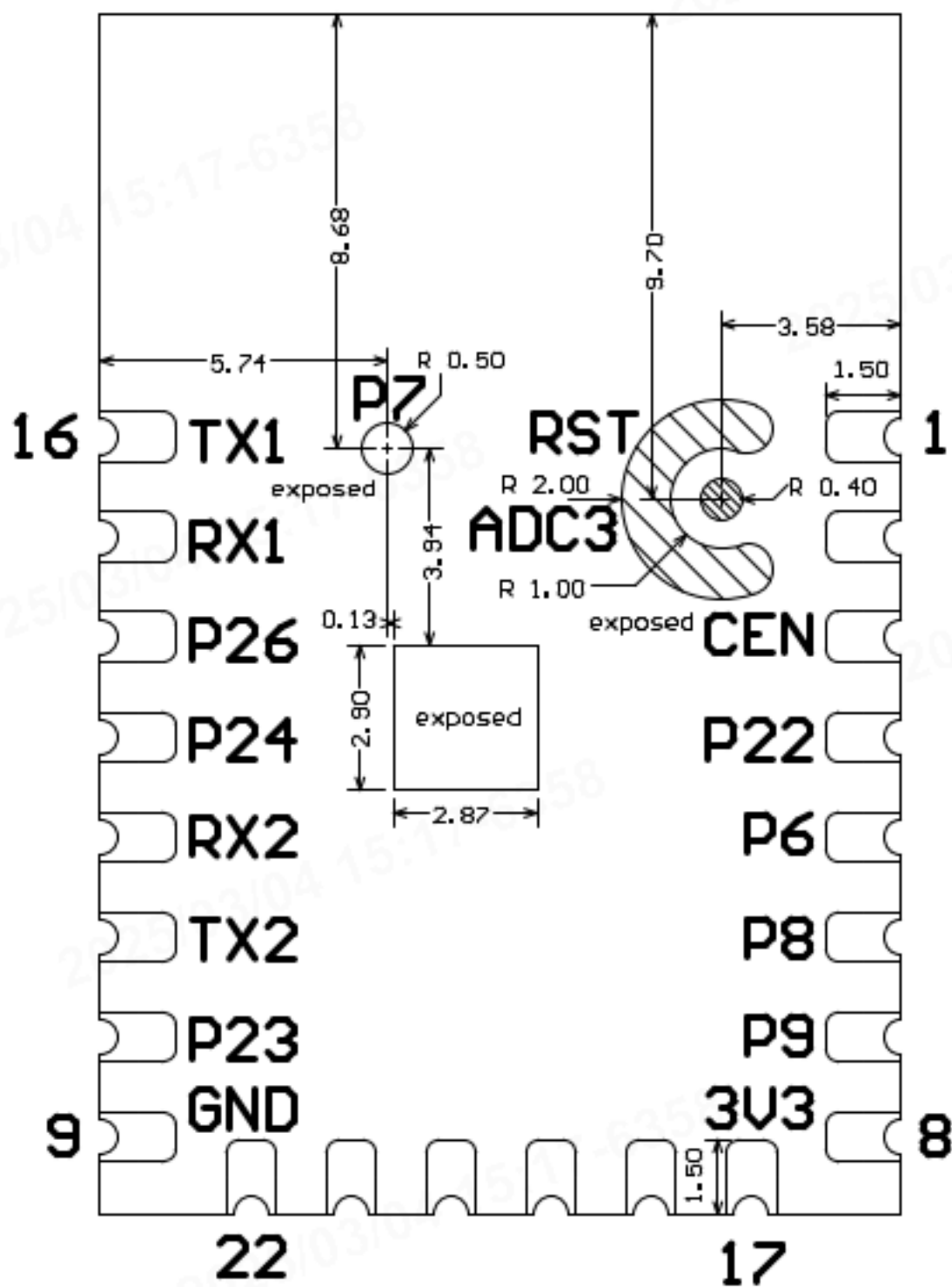
Make sure that the enclosure surrounding the antenna is not traced or filled with copper. Otherwise, the RF performance might be degraded.

6. Packing and production instructions

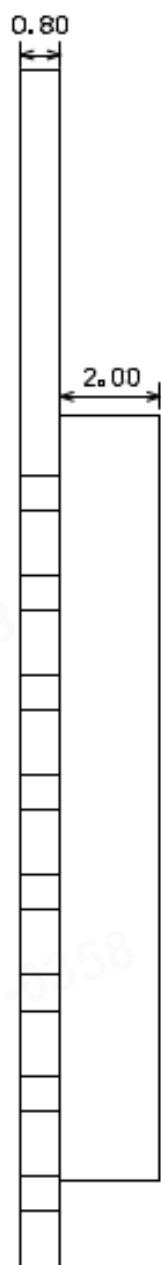
6.1. Mechanical dimensions

Dimensions of the T1-3S PCB are 16.0 ± 0.35 mm (W) $\times$ 24.0 ± 0.35 mm (L) $\times$ 2.8 ± 0.15 mm (H).





Bottom View



Unit: mm

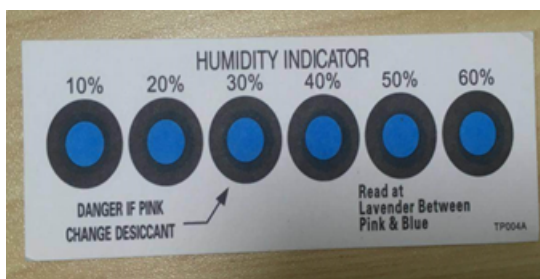
Module form factor tolerance: $\pm 0.35\text{mm}$

PCB thickness tolerance: $\pm 0.1\text{mm}$

Shield cover height tolerance: $\pm 0.05\text{mm}$

The figure below shows the recommended footprint of the T1-3S module.

- SPI
- Reflow soldering machine
- Thermal profiler
- Automated optical inspection (AOI) device
- Instruments or devices required for the wave soldering process:
- Wave soldering device
- Wave soldering fixture
- Constant-temperature soldering iron
- Tin bar, tin wire, and flux
- Thermal profiler
- Instruments or devices required for the baking process:
- Cabinet oven
- Electrostatic discharge (ESD) protection and heat-resistant trays
- ESD protection and heat-resistant gloves
- A delivered module must meet the following storage requirements:
- The moisture-proof bag must be placed in an environment where the temperature is below 40°C and the relative humidity is lower than 90%.
- The shelf life of a dry-packaged product is 12 months from the date when the product is packaged and sealed.
- A humidity indicator card (HIC) is put in the sealed package.



- The module needs to be baked in the following cases:
- The vacuum packaging bag is damaged before unpacking.
- After unpacking, no HIC is found in the packaging bag.
- After unpacking, the HIC indicates a humidity level of 10% or higher. In this case, the circle turns pink on the HIC.
- The total exposure time has lasted for over 168 hours since unpacking.
- More than 12 months have passed since the first sealing of the bag.

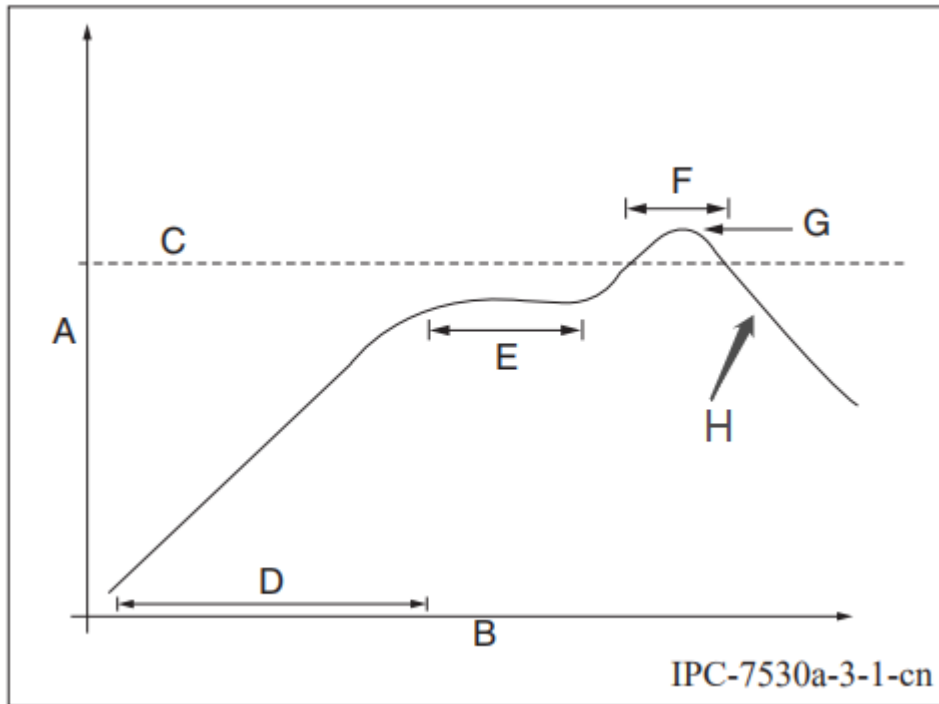
- The baking parameter settings are described below:
- Baking temperature: 40°C for reel packaging with relative humidity $\leq$ 5%. And 125°C for tray packaging with relative humidity $\leq$ 5% (use the heat-resistant tray, rather than plastic containers).
- Baking time: 168 hours for reel packaging and 12 hours for tray packaging.
- Temperature for triggering an alert: 50°C for reel packaging and 135°C for tray packaging.
- Production can begin after a module has cooled down to below 36°C under natural conditions.
- If a module remains unused for over 168 hours after being baked, it needs to be baked again.
- If a batch of modules is not baked after exposure for more than 168 hours, do not use reflow soldering or wave soldering to solder them. Because these modules are level-3 moisture-sensitive devices, they are very likely to get damp when exposed beyond the allowable time. In this case, if they are soldered at high temperatures, device failure or poor soldering performance might occur.
- In the whole production process, take electrostatic discharge (ESD) protective measures.
- To guarantee the pass rate, we recommend that you use the SPI and AOI to monitor the quality of solder paste printing and mounting.

6.3. Recommended oven temperature curve

Select a proper soldering technique according to the process. For more information, refer to the recommended oven temperature curve of either reflow soldering or wave soldering. The set temperatures might deviate from the actual temperature measurements. All temperatures shown in this module datasheet are obtained through actual measurements.

Technique 1: SMT process (recommended oven temperature curve of reflow soldering)

Set the oven temperatures according to the following curve.



- A: temperature axis
- B: time axis
- C: alloy liquidus temperature from 217°C to 220°C
- D: ramp-up slope from 1°C/s to 3°C/s
- E: keep a constant temperature from 150°C to 200°C for a time period of 60s to 120s
- F: temperature above liquidus temperature for 50s to 70s
- G: peak temperature from 235°C to 245°C
- H: ramp-down slope from 1°C/s to 4°C/s

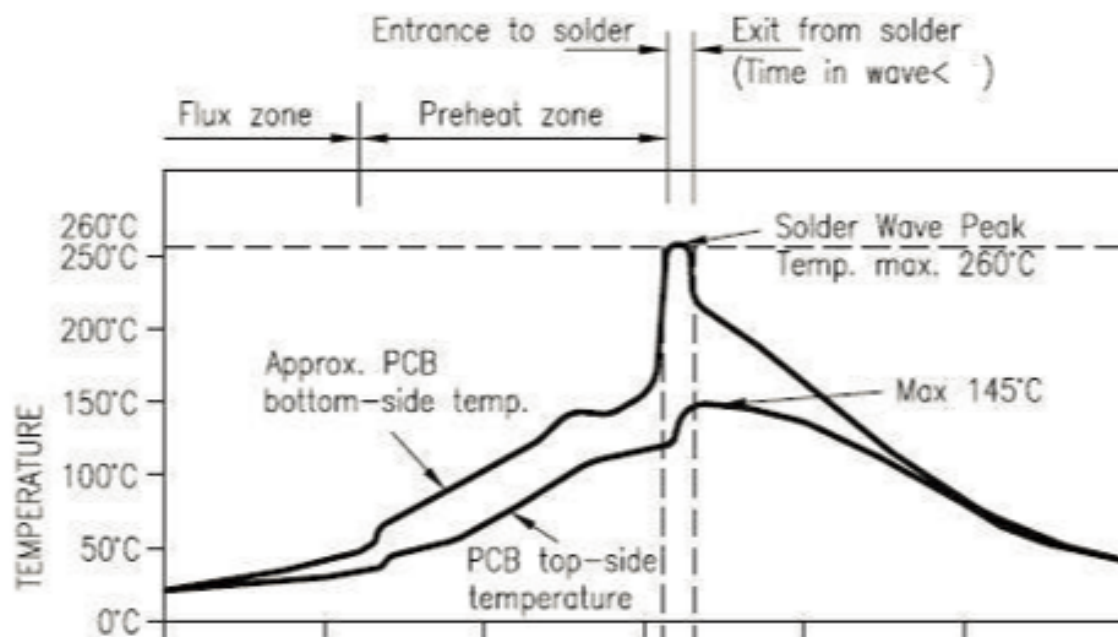


The curve above is based on solder paste SAC305. For more information about other solder pastes, see the recommended oven temperature curve in the specified solder paste specifications.

Technique 2: Wave soldering process (oven temperature curve of wave soldering)

Set the oven temperatures according to the following temperature curve of wave soldering. The peak temperature is $260^{\circ}\text{C} \pm 5^{\circ}\text{C}$.

DIP Type Product Pass Wavesolder Graph



Suggestions on wave soldering

Suggestions on manual repair soldering

Preheat temperature	80°C to 130°C	Soldering temperature	360°C ± 20°C
Preheat duration	75s to 100s	Soldering duration	Less than 3s/point
Contact duration at the peak	3s to 5s	N/A	N/A
Solder tank temperature	260°C ± 5°C	N/A	N/A
Ramp-up slope	≤ 2°C/s	N/A	N/A
Ramp-down slope	≤ 6°C/s	N/A	N/A

6.4. Storage conditions

	Caution This bag contains MOISTURE-SENSITIVE DEVICES	LEVEL 3 If blank, see adjacent bar code label
1. Calculated shelf life in sealed bag: 12 months at <40°C and <90% relative humidity (RH)		
2. Peak package body temperature: <u>260</u> °C If blank, see adjacent bar code label		
3. After bag is opened, devices that will be subjected to reflow solder or other high temperature process must be		
a) Mounted within: <u>168</u> hours of factory conditions If blank, see adjacent bar code label ≤30°C/60% RH, or		
b) Stored per J-STD-033		
4. Devices require bake, before mounting, if:		
a) Humidity Indicator Card reads >10% for level 2a - 5a devices or >60% for level 2 devices when read at 23 ± 5°C		
b) 3a or 3b are not met		
5. If baking is required, refer to IPC/JEDEC J-STD-033 for bake procedure		
Bag Seal Date: <u>See Production Date</u> If blank, see adjacent bar code label		
Note: Level and body temperature defined by IPC/JEDEC J-STD-020		

7. MOQ and packaging information

Product model	MOQ (pcs)	Shipping packaging	Modules per reel	Reels per carton
T1-3S	3600	Tape and reel	900	4

8. Appendix: Statement

FCC Caution: Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate this device. This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

List of applicable FCC rules

This module has been granted modular approval as below listed FCC rule parts. - FCC Rule parts 15C (15.247)

This device generates, uses, and can radiate radio frequency energy and, if not installed and used following the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this device does cause harmful interference to radio or television reception, which can be determined by turning the device off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the device and receiver.
- Connect the device to an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Radiation Exposure Statement

This device complies with FCC radiation exposure limits set forth for an uncontrolled rolled environment. This device should be installed and operated with a minimum distance of 20cm between the radiator and your body.

Important Note

This radio module must not be installed to co-locate and operate simultaneously with other radios in the host system except by following FCC multi-transmitter product procedures. Additional testing and device authorization may be required to operate simultaneously with other radios.

The availability of some specific channels and/or operational frequency bands are country dependent and are firmware programmed at the factory to match the intended destination. The firmware setting is not accessible to the end-user.

The host product manufacturer is responsible for compliance with any other FCC rules that apply to the host not covered by the modular transmitter grant of certification. The final host product still requires Part 15 Subpart B compliance testing with the modular transmitter installed. The separate approval is required for all other operating configurations including portable configurations with respect to Part 2.1093 and different antenna configurations.

The end-user manual shall include all required regulatory information/warnings as shown in this manual, including “This product must be installed and operated with a minimum distance of 20 cm between the radiator and user body”. The OEM integrator is responsible for ensuring that the end-user has no manual instructions to remove or install the module.

This device has got an FCC ID: 2ANDL-T1-3S. The end product must be labeled in a visible area with the following: “Contains Transmitter Module FCC ID: 2ANDL-T1-3S”.

This device is intended only for OEM integrators under the following conditions: The antenna must be installed such that 20cm is maintained between the antenna and users, and the transmitter module may not be co-located with any other transmitter or antenna.

As long as the 2 conditions above are met, further transmitter tests will not be required. However, the OEM integrator is still responsible for testing their end-product for any additional compliance requirements required with this module installed.

Declaration of Conformity European Notice



Hereby, Hangzhou Tuya Information Technology Co., Ltd declares that this module product is in compliance with essential requirements and other relevant provisions of Directive 2014/53/EU, 2011/65/EU. A copy of the Declaration of Conformity can be found at <https://www.tuya.com>.



This product must not be disposed of as normal household waste, in accordance with the EU directive for waste electrical and electronic equipment (WEEE-2012/19/EU). Instead, it should be disposed of by returning it to the point of sale, or to a municipal recycling collection point.

The device could be used with a separation distance of 20cm from the human body.

Industry Canada Statement This device complies with Industry Canada's licence-exempt RSSs. Operation is subject to the following two conditions: (1) This device may not cause interference. (2) This device must accept any interference, including interference that may cause undesired operation of the device. Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes: (1) l'appareil ne doit pas produire de brouillage. (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

Radiation Exposure Statement This equipment complies with IC radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with minimum distance 20 cm between the radiator & your body.

Déclaration d'exposition aux radiations: Cet équipement est conforme aux limites d'exposition aux rayonnements ISED établies pour un environnement non contrôlé. Cet équipement doit être installé et utilisé avec un minimum de 20 cm de distance entre la source de rayonnement et votre corps. L'appareil peut interrompre automatiquement la transmission en cas d'absence d'informations à transmettre ou de panne opérationnelle. Notez que ceci n'est pas destiné à

interdire la transmission d'informations de contrôle ou de signalisation ou l'utilisation de codes répétitifs lorsque cela est requis par la technologie.

This device is intended only for OEM integrators under the following

conditions: (1) The antenna must be installed such that 20 cm is maintained between the antenna and users. (2) The transmitter module may not be co-located with any other transmitter or antenna. As long as 2 conditions above are met, further transmitter test will not be required. However, the OEM integrator is still responsible for testing their end-product for any additional compliance requirements required with this module installed.

Cet appareil est conçu uniquement pour les intégrateurs OEM dans les conditions suivantes: (Pour utilisation de dispositif module) (1) L'antenne doit être installée de telle sorte qu'une distance de 20 cm est respectée entre l'antenne et les utilisateurs. (2) Le module émetteur peut ne pas être coïmplanté avec un autre é metteur ou antenne.

Tant que les 2 conditions ci-dessus sont remplies, des essais supplé mentaires sur l'émetteur nésont pas nécessaires. Toutefois, l'intégrateur OEM est toujours responsable des essais sur son produit final pour toutes exigences de conformité supplémentaires requis pour ce module installé.

IMPORTANT NOTE: In the event that these conditions can not be met (for example certain laptop configurations or colocation with another transmitter), then the Canada authorization is no longer considered valid and the IC ID can not be used on the final product. In these circumstances, the OEM integrator will be responsible for re-evaluating the end product (including the transmitter) and obtaining a separate Canada authorization.

NOTE IMPORTANTE: Dans le cas où ces conditions ne peuvent être satisfaites (par exemple pour certaines configurations d'ordinateur portable ou de certaines co-localisation avec un autre émetteur), l'autorisation du Canada n'est plus considéré comme valide et l'ID IC ne peut pas être utilisé sur le produit final. Dans ces circonstances, l'intégrateur OEM sera chargé de réévaluer le produit final (y compris l'émetteur) et l'obtention d'une autorisation distincte au Canada.

End Product Labeling This transmitter module is authorized only for use in device where the antenna may be installed such that 20 cm may be maintained

between the antenna and users. The final end product must be labeled in a visible area with the following: "Contains IC:23243-T13S" .

Plaque signalétique du produit final Ce module émetteur est autorisé uniquement pour une utilisation dans un dispositif où l'antenne peut être installée de telle sorte qu'une distance de 20cm peut être maintenue entre l'antenne et les utilisateurs. Le produit final doit être étiqueté dans un endroit visible avec l'inscription suivante: "Contient des IC:23243-T13S".

Manual Information To the End User The OEM integrator has to be aware not to provide information to the end user regarding how to install or remove this RF module in the user's manual of the end product which integrates this module. The end user manual shall include all required regulatory information/warning as show in this manual.

Manuel d'information à l'utilisateur final L'intégrateur OEM doit être conscient de ne pas fournir des informations à l'utilisateur final quant à la façon d'installer ou de supprimer ce module RF dans le manuel de l'utilisateur du produit final qui intègre ce module. Le manuel de l'utilisateur final doit inclure toutes les informations réglementaires requises et avertissements comme indiqué dans ce manuel.